LOYOLA
UNIVERSITY MARYLAND

Economics 220: Business Statistics
Fall Semester 2026
Maryland Hall 340
Tuesdays and Thursdays 10:50AM - 12:05PM

Instructor: Dr. Kerry Tan
Email: kmtan@loyola.edu
Zoom: https://loyola.zoom.us/my/kmtan
Office: Sellinger Hall 321
Office Hours: By appointment (Calendly)

Course Website: https://moodle.loyola.edu/
Lecture slides, Zoom recordings, homework, and other course materials are posted on Moodle.

Course Overview:
This course introduces you to the concepts and applications of statistics used in business. There will be significant emphasis on the use of Microsoft Excel, a popular spreadsheet software. You will learn how to use Excel to illustrate data, calculate descriptive statistics, run regressions, and interpret the results. By the end of this course, you should confidently include your experience with Excel as a skill on your resume and LinkedIn profile, which will help you be more competitive on the job market.

The course objectives are for students to:

- Demonstrate competency in Excel, particularly in the statistical tools in Data Analysis.
- Understand the basics of business statistics including data visualizations, descriptive statistics, probability theory, hypothesis testing, and regression estimation.
- Be able to successfully apply these techniques to multiple variable situations found in business disciplines.

Textbook:
David Anderson, Dennis Sweeney, Thomas Williams, Jeffrey Camm, and James Cochran
Essentials of Modern Business Statistics with Microsoft Office Excel (7th Edition), Cengage (2018)
ISBN: 9781337298292

Based on my conversation with previous students, it seems as though you can successfully complete this course without ever opening the textbook as long as you attend class regularly and ask me questions whenever you need clarification. As such, you are not required to purchase the textbook.

Microsoft Excel Analysis ToolPak:
The Analysis ToolPak is an Excel add-in that provides data analysis tools. To load the Analysis ToolPak add-in, execute the following steps:

1. On the Tools tab, click "Excel Add-ins..."
2. Check the box for "Analysis ToolPak" and click the OK button.

Grading Policy:
Grades for this course will be assigned as follows:
Exam 1: 25% (October 8)
Exam 2: 25% (November 5)
Exam 3: 25% (December 15)
Homework Assignments: 20%
Surveys: 5%

Exams:
There will be 3 in-class exams, each consisting of 15 multiple choice questions and 3 word problems. Exam 1 will only cover Chapters 1, 2, and 3. Exam 2 will only cover Chapters 4, 6, and 9. Exam 3 will only cover Chapters 14 and 15. The exams are closed book and closed notes; however, you are allowed to use one handwritten 3"x5" notecard.

Makeup exams will *not* be given without documented medical emergency or family emergency. Travel plans are *not* an emergency. It is your responsibility to plan around scheduled exam times.

Homework Assignments:
3 homework assignments will be assigned in this course. The homework (and solutions) will be posted on the course website. Each homework assignment will be worth 25 points and will be graded based on accuracy. Students are encouraged to collaborate in small groups, but each student must turn in his or her own assignment. Students must hand in their homework in class on the assigned due date. No late homework will be accepted without a valid excuse.

Surveys:
At the end of each week, you will fill out a survey on Moodle that provides feedback on your understanding of the course material. I will address your survey responses at the beginning of each class on the following Tuesday. Since surveys are intended to be low-effort/high-reward assignments, I do not give extra credit, especially at the end of the semester.

Grading Scale:
Generally, final letter grades will follow the typical structure of: A (93-100), A- (90-92.99), B+ (87-89.99), B (83-86.99), B- (80-82.99), C+ (77-79.99), C (73-76.99), C- (70-72.99), D+ (67-69.99), D (60-66.99), and F (below 60). However, this breakdown is not set in stone and can vary according to the distribution of raw grades.

Students with Disabilities:
If you are a student with a disability requesting accommodations for this course, please submit your Faculty Notification Email (FNE) request to Disability Support Services as soon as possible. After the FNE has been sent, please schedule a brief meeting with me during my office hours to discuss accommodations. If you need additional academic accommodations because of remote learning, please contact your DSS advisor immediately.

If you are not registered with DSS, but feel you might need accommodations or modifications for my course because of a physical, medical, emotional, or learning impairment, you must register with Disability Support Services. To register with DSS, or if you have questions about disability accommodations, please contact the Disability Support Services Office at dss@loyola.edu.

Academic Misconduct:
All students are expected to understand the meaning of the Loyola Honor Code, which states:

"All students of the Loyola Community have been equally entrusted by their peers to conduct themselves honestly on all academic assignments.

The students of this University understand that having collective and individual responsibility for the ethical welfare of their peers exemplifies a commitment to the community. Students who submit materials that are the products of their own minds demonstrate respect for themselves and the community in which they study."

Ignorance of the Code is not a valid reason for committing an act of academic dishonesty. The following constitute violations of the Code: cheating, stealing, lying, forgery, plagiarism, and the failure to report a violation. Violations of the Honor Code will be handled by the Honor Council.

Course Schedule:

Date	**Chapter Covered**
Sept. 1, 3	Syllabus/Ch 1 - Data and Statistics
Sept. 8, 10, 15, 17	Ch 2 - Descriptive Statistics: Tabular and Graphical Displays
Sept. 22, 24, 29, Oct. 1	Ch 3 - Descriptive Statistics: Numerical Measures
Oct. 6	Review for Exam 1
Oct. 8	EXAM 1
Oct. 13, 15, 20	Ch 4 - Introduction to Probability & Ch 6 - Continuous Probability Distributions
Oct. 22, 27, 29	Ch 9 - Hypothesis Tests
Nov. 3	Review for Exam 2
Nov. 5	EXAM 2
Nov. 10, 12, 17	Ch 14 - Simple Linear Regression
Nov. 19, Dec. 1, 3	Ch 15 - Multiple Regression
Nov. 24, 26	No Class: Thanksgiving Break
Dec. 8	Review for Exam 3
Dec. 15 9:00AM - 10:15AM	EXAM 3